

Wednesday, 24 December 2025

# DevOps-focused Linux command reference — the commands most commonly used in real DevOps

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This is the kind of list used by Linux admins, SREs, Cloud & DevOps engineers daily.

## 1. FILE & DIRECTORY MANAGEMENT

### **ls**

List directory contents

```
ls -l  
ls -a  
cd
```

Change directory

```
cd /var/log  
pwd
```

Print current directory

```
pwd
```

### **mkdir**

Create directory

```
mkdir app  
mkdir -p app/logs
```

### **rmdir**

Remove empty directory

```
rmdir test
```

## **rm**

Delete files/directories

```
rm file.txt
rm -rf folder
```

## **cp**

Copy files

```
cp file1 file2
cp -r dir1 dir2
```

## **mv**

Move or rename

```
mv old.txt new.txt
```

## **touch**

Create empty file

```
touch app.log
```

# **2. FILE VIEWING & EDITING**

## **cat**

Display file content

```
cat file.txt
```

## **less**

Scrollable file viewer

```
less large.log
```

## **more**

Paged file output

**more file.txt**

## **head**

First lines of file

**head -n 10 file.txt**

## **tail**

Last lines (log monitoring)

**tail -f app.log**

## **nano, vi, vim**

File editors

**vi config.yaml**

# **3. SEARCH & TEXT PROCESSING (VERY IMPORTANT)**

## **grep**

Search text

**grep "ERROR" app.log**  
**grep -i error app.log**

## **awk**

Text processing

**awk '{print \$1}' file.txt**

## **sed**

Stream editor

**sed 's/old/new/g' file.txt**

## **cut**

Extract columns

```
cut -d ":" -f1 /etc/passwd
```

## **sort**

Sort output

```
sort file.txt
```

## **uniq**

Remove duplicates

```
uniq file.txt
```

## **wc**

Count lines/words

```
wc -l file.txt
```

# **4. FILE PERMISSIONS & OWNERSHIP**

## **chmod**

Change permissions

```
chmod 755 script.sh
```

## **chown**

Change owner

```
chown user:group file.txt
```

## **chgrp**

Change group

```
chgrp devops file.txt
```

## 5. USER & GROUP MANAGEMENT

### **useradd**

Create user

**useradd devuser**

### **passwd**

Set password

**passwd devuser**

### **userdel**

Delete user

**userdel devuser**

### **groupadd**

Create group

**groupadd devops**

### **groups**

Check user groups

**groups devuser**

## 6. PROCESS MANAGEMENT

### **ps**

Process status

**ps aux**

## **top**

Real-time process view

```
top
```

## **htop**

Advanced process viewer

```
htop
```

## **kill**

Kill process

```
kill 1234
```

## **killall**

Kill by name

```
killall nginx
```

## **nice**

Set priority

```
nice -n 10 command
```

# **7. SYSTEM MONITORING**

## **df**

Disk usage

```
df -h
```

## **du**

Directory size

```
du -sh /var/log
```

## **free**

Memory usage

**free -m**

## **uptime**

System run time

**uptime**

## **vmstat**

Memory statistics

**vmstat**

## **iostat**

Disk IO stats

**iostat**

# **8. NETWORKING COMMANDS**

## **ip**

Network configuration

**ip addr**  
**ip route**

## **ifconfig**

Legacy network tool

**ifconfig**

## **ping**

Test connectivity

**ping [google.com](http://google.com)**

## **netstat**

Network connections

**netstat -tulnp**

## **ss**

Modern netstat replacement

**ss -tuln**

## **curl**

API testing / HTTP calls

**curl <http://example.com>**

## **wget**

Download files

**wget <http://file.com/app.tar.gz>**

# **9. PACKAGE MANAGEMENT**

## **Debian/Ubuntu**

**apt update**  
**apt install nginx**  
**apt remove nginx**

## **RHEL/CentOS**

**yum install nginx**  
**dnf install nginx**

## 10. ARCHIVE & COMPRESSION

### **tar**

Create/extract archives

```
tar -cvf app.tar app/  
tar -xvf app.tar
```

### **gzip**

Compress

```
gzip file.txt
```

### **zip**

Zip files

```
zip app.zip file.txt
```

## 11. ENVIRONMENT VARIABLES

### **export**

Set variable

```
export APP_ENV=prod
```

### **env**

List variables

```
env
```

### **printenv**

Print variable

```
printenv PATH
```

## 12. SYSTEM SERVICES (SYSTEMD)

### **systemctl**

Manage services

```
systemctl start nginx
systemctl stop nginx
systemctl status nginx
systemctl enable nginx
```

### **journalctl**

View logs

```
journalctl -u nginx
```

## 13. DISK & FILESYSTEM

### **mount**

Mount filesystem

```
mount /dev/sdb1 /mnt
```

### **umount**

Unmount

```
umount /mnt
```

### **lsblk**

Block devices

```
lsblk
```

## 14. SSH & REMOTE ACCESS (CRITICAL FOR DEVOPS)

### **ssh**

Remote login

```
ssh user@server
```

### **scp**

Secure copy

```
scp file.txt user@server:/path
```

### **rsync**

Sync files

```
rsync -av app/ server:/app/
```

## 15. SHELL & SCRIPTING

### **bash**

Run shell

```
bash script.sh
```

### **sh**

Shell interpreter

```
sh script.sh
```

### **source**

Load environment

```
source ~/.bashrc
```

## 16. DEVOPS-SPECIFIC DAILY COMMANDS

| Command                       | Use             |
|-------------------------------|-----------------|
| <code>docker ps</code>        | List containers |
| <code>docker logs</code>      | Container logs  |
| <code>kubectl get pods</code> | Kubernetes pods |
| <code>kubectl logs</code>     | Pod logs        |
| <code>crontab -l</code>       | Scheduled jobs  |
| <code>crontab -e</code>       | Edit cron       |

### HOW DEVOPS ENGINEERS USE THESE DAILY

- Debug production issues (`ps`, `top`, `tail`)
- Deploy apps (`scp`, `rsync`, `tar`)
- Monitor servers (`df`, `free`, `uptime`)
- Manage services (`systemctl`)
- Troubleshoot networking (`curl`, `ping`)
- Automate with scripts (`bash`, `cron`)

# MOST IMPORTANT LINUX COMMANDS FOR INTERVIEWS

## **1** FILE & DIRECTORY COMMANDS (VERY COMMON)

### **ls**

List files

```
ls -l
ls -a
ls -lh
```

**Interview tip:** Difference between **-a** and **-l**.

### **cd**

Change directory

```
cd /var/log
cd ..
```

### **pwd**

Current directory

```
pwd
```

### **mkdir**

Create directory

```
mkdir app
mkdir -p app/logs
```

**Q:** What does **-p** do?

### **rm**

Delete files/directories

```
rm file.txt
rm -rf folder
```

⚠ Interview warning: **Explain why `rm -rf /` is dangerous.**

## **cp**

Copy files

```
cp file1 file2
cp -r dir1 dir2
```

## **mv**

Move or rename

```
mv old.txt new.txt
```

# **2 FILE VIEWING COMMANDS (LOG ANALYSIS)**

## **cat**

View file

```
cat file.txt
```

## **less**

Scrollable view

```
less app.log
```

**Q: Difference between `cat` and `less`?**

## **head**

First lines

```
head -n 10 file.txt
```

## **tail**

Last lines (logs)

```
tail -f app.log
```

**Q: What does -f do?**

## **3 SEARCH & TEXT PROCESSING (HIGHLY ASKED)**

### **grep**

Search text

```
grep "ERROR" app.log  
grep -i error app.log  
grep -r "port" /etc/
```

### **awk**

Column processing

```
awk '{print $1}' file.txt
```

**Q: When to use awk vs cut?**

### **sed**

Replace text

```
sed 's/old/new/g' file.txt
```

### **cut**

Extract fields

```
cut -d ":" -f1 /etc/passwd
```

### **sort**

Sort output

```
sort file.txt
```

## **uniq**

Remove duplicates

```
uniq file.txt
```

Q: Why sort before uniq?

## **4 FILE PERMISSIONS & OWNERSHIP (VERY IMPORTANT)**

### **chmod**

Change permissions

```
chmod 755 script.sh  
chmod u+x file.sh
```

Interview MUST know:

```
r = 4, w = 2, x = 1
```

### **chown**

Change owner

```
chown user:group file.txt
```

### **umask**

Default permissions

```
umask
```

## **5 PROCESS MANAGEMENT (FREQUENT)**

### **ps**

Process status

```
ps aux
```

## **top**

Live processes

**top**

## **htop**

Enhanced top

**htop**

## **kill**

Kill process

**kill 1234**

**kill -9 1234**

**Q:** Difference between **kill** and **kill -9**?

## **nice**

Priority

**nice -n 10 command**

## **SYSTEM MONITORING (REAL-WORLD QUESTIONS)**

### **df**

Disk usage

**df -h**

### **du**

Directory size

**du -sh /var/log**

### **free**

Memory

```
free -m
```

**uptime**

System run time

```
uptime
```

## NETWORKING COMMANDS (DEVOPS FAVORITE)

**ip**

Network info

```
ip addr  
ip route
```

**ping**

Connectivity test

```
ping google.com
```

**netstat**

Ports & connections

```
netstat -tulnp
```

**ss**

Modern netstat

```
ss -tuln
```

## **curl**

API testing

```
curl http://example.com  
curl -I example.com
```

## **wget**

Download files

```
wget http://file.com/app.tar.gz
```

## **8 SERVICE MANAGEMENT (SYSTEMD)**

### **systemctl**

Manage services

```
systemctl start nginx  
systemctl stop nginx  
systemctl restart nginx  
systemctl status nginx  
systemctl enable nginx
```

### **journalctl**

Logs

```
journalctl -u nginx
```

## **9 SSH & REMOTE ACCESS (VERY COMMON)**

### **ssh**

Remote login

```
ssh user@server
```

## **scp**

Copy files remotely

```
scp file.txt user@server:/path
```

## **rsync**

Fast sync

```
rsync -av app/ server:/app/
```

# **10 ARCHIVE & COMPRESSION**

## **tar**

Archive

```
tar -cvf app.tar app/  
tar -xvf app.tar
```

## **gzip**

Compress

```
gzip file.txt
```

# **11 ENVIRONMENT VARIABLES**

## **export**

Set variable

```
export ENV=prod
```

## env

List variables

env

## 1 2 CRON JOBS (AUTOMATION)

### crontab

Schedule jobs

crontab -l

crontab -e

## 1 3 DEVOPS INTERVIEW BONUS COMMANDS

| Command          | Purpose            |
|------------------|--------------------|
| docker ps        | Running containers |
| docker logs      | Container logs     |
| kubectl get pods | Kubernetes pods    |
| kubectl logs pod | Pod logs           |
| history          | Command history    |
| alias            | Command shortcut   |

## TOP INTERVIEW QUESTIONS BASED ON COMMANDS

- ✓ Difference between **grep, awk, sed**
- ✓ How to find large files in Linux
- ✓ How to check which port is used
- ✓ How to troubleshoot high CPU
- ✓ How to monitor logs live
- ✓ How to find zombie processes

# 50 LINUX INTERVIEW QUESTIONS & ANSWERS

## 1. What is Linux?

Linux is an **open-source, Unix-like operating system kernel** used in servers, cloud, and embedded systems.

## 2. What is the Linux kernel?

The kernel is the **core of Linux** that manages **CPU, memory, devices, and system calls**.

## 3. Difference between Linux and Unix?

Linux is **open-source**; Unix is mostly **proprietary**.

## 4. What is a shell?

A shell is a **command-line interface** between the user and the kernel (e.g., bash).

## 5. What is root user?

The root user has **full administrative privileges**.

## 6. What is / in Linux?

The **root directory**, the top of the filesystem hierarchy.

## 7. What is /home?

Stores **user home directories**.

## 8. What is /etc?

Contains **system configuration files**.

## 9. What is /var?

Stores **variable data** like logs, caches.

## 10. What is **/bin** and **/sbin**?

Essential **binary commands**; **/sbin** mainly for system admin.

## ◆ **FILE & DIRECTORY COMMANDS (11–20)**

### 11. Difference between **ls -l** and **ls -a**?

**-l** shows details; **-a** shows hidden files.

### 12. What does **pwd** do?

Prints **current working directory**.

### 13. What is the use of **mkdir -p**?

Creates **parent directories** automatically.

### 14. Difference between **rm -r** and **rm -f**?

**-r** deletes directories recursively; **-f** forces deletion.

### 15. What is **cp -r**?

Copies directories **recursively**.

### 16. What does **touch** do?

Creates an **empty file** or updates timestamp.

### 17. Difference between **cat** and **less**?

**cat** prints all content; **less** allows **scrolling**.

### 18. How to view last 10 lines of a file?

```
tail -n 10 file
```

### 19. What does **tail -f** do?

Displays **live log updates**.

### 20. How to find a file in Linux?

```
find / -name file.txt
```

## ◆ PERMISSIONS & OWNERSHIP (21–30)

### 21. What are Linux file permissions?

Read (r), Write (w), Execute (x).

### 22. Explain permission number 755.

Owner: rwx, Group: r-x, Others: r-x.

### 23. What does **chmod** do?

Changes file permissions.

### 24. What does **chown** do?

Changes file **owner and group**.

### 25. What is **umask**?

Default permission mask.

### 26. Difference between hard link and soft link?

Hard link points to inode; soft link points to file path.

## 27. What is inode?

Stores **file metadata** (not filename).

## 28. How to check file permissions?

```
ls -l
```

## 29. How to give execute permission?

```
chmod +x file.sh
```

## 30. What is sticky bit?

Prevents users from deleting others' files (e.g. /tmp).

# ◆ PROCESS & SYSTEM MANAGEMENT (31–40)

## 31. What is a process?

A running instance of a program.

## 32. Difference between process and thread?

Threads share memory; processes don't.

## 33. How to check running processes?

```
ps aux
```

## 34. What is **top**?

Shows **real-time system processes**.

## 35. What does **kill -9** do?

Forcefully terminates a process.

### 36. What is load average?

Average number of processes waiting for CPU.

### 37. How to check memory usage?

```
free -m
```

### 38. How to check disk usage?

```
df -h
```

### 39. What is swap memory?

Disk space used as **virtual RAM**.

### 40. How to check system uptime?

```
uptime
```

## ◆ NETWORKING & SERVICES (41–50)

### 41. How to check IP address?

```
ip addr
```

### 42. What does **ping** do?

Tests **network connectivity**.

### 43. Difference between **netstat** and **ss**?

**ss** is faster and modern replacement.

### 44. How to check open ports?

```
ss -tulnp
```

#### **45. What is `curl` used for?**

Testing APIs and HTTP requests.

#### **46. What is SSH?**

Secure remote login protocol.

#### **47. What is `systemctl`?**

Manages services in systemd.

#### **48. How to start a service?**

```
systemctl start nginx
```

#### **49. What is a cron job?**

Scheduled task automation.

#### **50. How to view cron jobs?**

```
crontab -l
```

**29th Dec**

### **Grep command**

**"`grep` is a Linux command used to [search for text patterns](#) inside files or command output."**

